

If s is the position function of P , then $s'(t) = v(t)$; that is, $s(t)$ is an antiderivative of $v(t)$. Hence, by the fundamental theorem of calculus,

$$\int_a^b v(t) dt = \int_a^b s'(t) dt = s(t) \Big|_a^b = s(b) - s(a).$$

Substituting in the formula for v_{av} gives us

$$v_{av} = \frac{s(b) - s(a)}{b - a},$$

which is the average velocity of P on $[a, b]$ (see (3.2)).

Results similar to that in Example 9 occur in discussions of average acceleration, average marginal cost, average marginal revenue, and many other applications of the derivative (see Exercises 45–50).

EXERCISES 5.6

Exer. 1–36: Evaluate the integral.

- 1 $\int_1^4 (x^2 - 4x - 3) dx$
- 2 $\int_{-2}^3 (5 + x - 6x^2) dx$
- 3 $\int_{-2}^3 (8z^3 + 3z - 1) dz$
- 4 $\int_0^2 (z^4 - 2z^3) dz$
- 5 $\int_7^{12} dx$
- 6 $\int_{-6}^{-1} 8 dx$
- 7 $\int_1^2 \frac{5}{x^6} dx$
- 8 $\int_1^4 \sqrt{16x^5} dx$
- 9 $\int_4^9 \frac{t-3}{\sqrt{t}} dt$
- 10 $\int_{-1}^{-2} \frac{2t-7}{t^3} dt$
- 11 $\int_{-8}^8 (\sqrt[3]{s^2} + 2) ds$
- 12 $\int_1^0 s^2 (\sqrt[3]{s} - \sqrt{s}) ds$
- 13 $\int_{-1}^0 (2x + 3)^2 dx$
- 14 $\int_1^2 (4x^{-5} - 5x^4) dx$
- 15 $\int_3^2 \frac{x^2 - 1}{x - 1} dx$
- 16 $\int_0^{-1} \frac{x^3 + 8}{x + 2} dx$
- 17 $\int_1^1 (4x^2 - 5)^{100} dx$
- 18 $\int_5^5 \sqrt[3]{x^2 + \sqrt{x^5 + 1}} dx$
- 19 $\int_1^3 \frac{2x^3 - 4x^2 + 5}{x^2} dx$
- 20 $\int_{-2}^{-1} \left(x - \frac{1}{x}\right)^2 dx$
- 21 $\int_{-3}^6 |x - 4| dx$
- 22 $\int_{-1}^5 |2x - 3| dx$
- 23 $\int_1^4 \sqrt{5 - x} dx$
- 24 $\int_1^5 \sqrt[3]{2x - 1} dx$
- 25 $\int_{-1}^1 (v^2 - 1)^3 v dv$
- 26 $\int_{-2}^0 \frac{v^2}{(v^3 - 2)^2} dv$
- 27 $\int_0^1 \frac{1}{(3 - 2x)^2} dx$
- 28 $\int_0^4 \frac{x}{\sqrt{x^2 + 9}} dx$
- 29 $\int_1^4 \frac{1}{\sqrt{x}(\sqrt{x} + 1)^3} dx$
- 30 $\int_0^1 (3 - x^4)^3 x^3 dx$

- 31 $\int_{\pi/2}^{\pi} \cos\left(\frac{1}{3}x\right) dx$
- 32 $\int_0^{\pi/2} 3 \sin\left(\frac{1}{2}x\right) dx$
- 33 $\int_{\pi/4}^{\pi/3} (4 \sin 2\theta + 6 \cos 3\theta) d\theta$
- 34 $\int_{\pi/6}^{\pi/4} (1 - \cos 4\theta) d\theta$
- 35 $\int_{-\pi/6}^{\pi/6} (x + \sin 5x) dx$
- 36 $\int_0^{\pi/3} \frac{\sin x}{\cos^2 x} dx$

Exer. 37–40: (a) Find a number z that satisfies the conclusion of the mean value theorem (5.28) for the given integral $\int_a^b f(x) dx$. (b) Find the average value of f on $[a, b]$.

- 37 $\int_0^4 \frac{x}{\sqrt{x^2 + 9}} dx$
- 38 $\int_{-2}^0 \sqrt[3]{x + 1} dx$
- 39 $\int_0^5 \sqrt{x + 4} dx$
- 40 $\int_{-3}^2 \sqrt{6 - x} dx$

Exer. 41–44: Find the derivative without integrating.

- 41 $D_x \int_0^3 \sqrt{x^2 + 16} dx$
- 42 $D_x \int_0^1 x \sqrt{x^2 + 4} dx$
- 43 $D_x \int_0^x \frac{1}{t + 1} dt$
- 44 $D_x \int_0^x \frac{1}{\sqrt{1 - t^2}} dt, \quad |x| < 1$

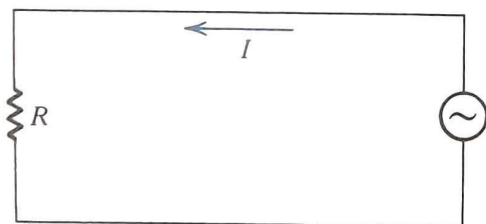
45 A point P is moving on a coordinate line with a continuous acceleration function a . If v is the velocity function, then the average acceleration on a time interval $[t_1, t_2]$ is

$$\frac{v(t_2) - v(t_1)}{t_2 - t_1}.$$

Show that the average acceleration is equal to the average value of a on $[t_1, t_2]$.

- 46 If a function f has a continuous derivative on $[a, b]$, show that the average rate of change of $f(x)$ with respect to x on $[a, b]$ (see Definition (3.4)) is equal to the average value of f' on $[a, b]$.
- 47 The vertical distribution of velocity of the water in a river may be approximated by $v = c(d - y)^{1/6}$, where v is the velocity (in m/sec) at a depth of y meters below the water surface, d is the depth of the river, and c is a positive constant.
- (a) Find a formula for the average velocity v_{av} in terms of d and c .
- (b) If v_0 is the velocity at the surface, show that $v_{av} = \frac{6}{7}v_0$.
- 48 In the electrical circuit shown in the figure, the alternating current I is given by $I = I_M \sin \omega t$, where t is the time and I_M is the maximum current. The rate P at which heat is being produced in the resistor of R ohms is given by $P = I^2 R$. Compute the average rate of production of heat over one complete cycle (from $t = 0$ to $t = 2\pi/\omega$). (Hint: Use the half-angle formula for the sine.)

EXERCISE 48



5.7 NUMERICAL INTEGRATION

To evaluate a definite integral $\int_a^b f(x) dx$ by means of the fundamental theorem of calculus, we need an antiderivative of f . If an antiderivative cannot be found, then we may use numerical methods to approximate the integral to any desired degree of accuracy. For example, if the norm of a partition of $[a, b]$ is small, then, by Definition (5.16), the definite integral can be approximated by any Riemann sum of f . In particular, if we use a regular partition with $\Delta x = (b - a)/n$, then

$$\int_a^b f(x) dx \approx \sum_{k=1}^n f(w_k) \Delta x,$$

where w_k is any number in the k th subinterval $[x_{k-1}, x_k]$ of the partition. Of course, the accuracy of the approximation depends on the nature of f and the magnitude of Δx . It may be necessary to make Δx very small to obtain a desired degree of accuracy. This, in turn, means that n is large, and hence the preceding sum contains many terms. Figure 5.5 illustrates the case in which $f(w_k)$ is the minimum value of f on $[x_{k-1}, x_k]$. In this case the error involved in the approximation is numerically the same as the area of the region that lies under the graph of f and over the inscribed rectangles.

- 49 If a ball is dropped from a height of s_0 feet above the ground and air resistance is negligible, then the distance it falls in t seconds is $16t^2$ feet. Use Definition (5.16) to show that the average velocity for the ball's journey to the ground is $4\sqrt{s_0}$ ft/sec.

- 50 A meteorologist determines that the temperature T ($^{\circ}\text{F}$) on a cold winter day is given by

$$T = \frac{1}{20}t(t - 12)(t - 24),$$

where t is time (in hours) and $t = 0$ corresponds to midnight. Find the average temperature between 6 A.M. and 12 noon.

- 51 If g is differentiable and f is continuous for every x , prove that

$$D_x \int_a^{g(x)} f(t) dt = f(g(x))g'(x).$$

- 52 Extend the formula in Exercise 51 to

$$D_x \int_{k(x)}^{g(x)} f(t) dt = f(g(x))g'(x) - f(k(x))k'(x).$$

Exer. 53–56: Use Exercises 51 and 52 to find the derivative.

$$53 \quad D_x \int_2^{x^4} \frac{t}{\sqrt{t^3 + 2}} dt$$

$$54 \quad D_x \int_0^{x^2} \sqrt{t^4 + 1} dt$$

$$55 \quad D_x \int_{3x}^{x^3} (t^3 + 1)^{10} dt$$

$$56 \quad D_x \int_{1/x}^{\sqrt{x}} \sqrt{t^4 + t^2 - 1} dt$$